



Comprehensive Compression Benchmark Results

Project: TAI - Algorithmic Information Theory

Date: 2026-03-19

Tools Compared: G07, gzip, bzip2, xz (LZMA), zstd

Table of Contents

- [1. Quick Summary](#)
- [2. Detailed Results by File](#)
- [3. Performance Metrics](#)
- [4. Rankings](#)
- [5. Overall Statistics](#)

Quick Summary

Compression Ratio Comparison

File	Size	G07	gzip	bzip2	xz	zstd	Winner
A	1.3MiB	53.40% ★	58.49%	54.05%	53.34% ★	53.52%	xz
B	1.2MiB	17.94% ★	22.89%	17.92% ★	17.76% ★	23.11%	xz
C	2.0MiB	26.67% ★	32.57%	26.51% ★	24.72% ★	31.19%	xz
D	2.0MiB	100.00%	100.02%	100.45%	100.01%	100.00%	g07

File	Size	G07	gzip	bzip2	xz	zstd	Winner
		★					
E	989KiB	85.60% ★	88.66%	88.43%	85.84%	88.53%	g07
F	2.0MiB	81.70% ★	88.01%	81.06% ★	82.38%	87.81%	bzip2
G	2.5MiB	28.85% ★	36.99%	28.70% ★	29.71%	35.82%	bzip2
H	999KiB	43.75% ★	44.02%	42.34% ★	35.88% ★	44.86%	xz

★ = Best compressor for that file

Detailed Results by File

File: A

Original Size: 1308765 bytes (1.3MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	683KiB (698834)	 53.40%	4.2717	88 ms	65	✓ 596KiB saved	✓
gzip	748KiB (765497)	 58.49%	4.6792	59 ms	16	✓ 531KiB saved	✓
bzip2	691KiB (707422)	 54.05%	4.3242	97 ms	69	✓ 588KiB	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
						saved	
xz	682KiB (698048)	53.34%	4.2669	362 ms	43	✓ 597KiB saved	✓
zstd	684KiB (700409)	53.52%	4.2813	25 ms	14	✓ 595KiB saved	✓

File: B

Original Size: 1168767 bytes (1.2MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	205KiB (209635)	17.94%	1.4349	68 ms	46	✓ 937KiB saved	✓
gzip	262KiB (267569)	22.89%	1.8315	50 ms	11	✓ 881KiB saved	✓
bzip2	205KiB (209479)	17.92%	1.4338	96 ms	38	✓ 937KiB saved	✓
xz	203KiB (207584)	17.76%	1.4209	276 ms	14	✓ 939KiB saved	✓
zstd	264KiB (270069)	23.11%	1.8486	11 ms	7	✓ 878KiB saved	✓

File: C

Original Size: 2000000 bytes (2.0MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	521KiB (533439)	26.67%	2.1338	56 ms	42	✓ 1.4MiB saved	✓
gzip	637KiB (651482)	32.57%	2.6059	64 ms	14	✓ 1.3MiB saved	✓
bzip2	518KiB (530253)	26.51%	2.1210	113 ms	70	✓ 1.5MiB saved	✓
xz	483KiB (494424)	24.72%	1.9777	501 ms	33	✓ 1.5MiB saved	✓
zstd	610KiB (623877)	31.19%	2.4955	28 ms	18	✓ 1.4MiB saved	✓

File: D

Original Size: 2000000 bytes (2.0MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	2.0MiB (2000009)	100.00%	8.0000	16 ms	14	✗ overhead	✓
gzip	2.0MiB (2000330)	100.02%	8.0013	65 ms	14	✗ overhead	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
bzip2	2.0MiB (2009059)	100.45%	8.0362	198 ms	112	✗ overhead	✓
xz	2.0MiB (2000168)	100.01%	8.0007	386 ms	11	✗ overhead	✓
zstd	2.0MiB (2000061)	100.00%	8.0002	16 ms	11	✗ overhead	✓

File: E

Original Size: 1012112 bytes (989KiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	847KiB (866338)	85.60%	6.8478	75 ms	193	✓ 143KiB saved	✓
gzip	877KiB (897332)	88.66%	7.0927	47 ms	12	✓ 113KiB saved	✓
bzip2	875KiB (895061)	88.43%	7.0748	107 ms	65	✓ 115KiB saved	✓
xz	849KiB (868764)	85.84%	6.8669	216 ms	48	✓ 140KiB saved	✓
zstd	875KiB (895975)	88.53%	7.0820	11 ms	14	✓ 114KiB saved	✓

File: F

Original Size: 2097152 bytes (2.0MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	1.7MiB (1713365)	81.70%	6.5360	105 ms	222	✓ 375KiB saved	✓
gzip	1.8MiB (1845625)	88.01%	7.0405	74 ms	21	✓ 246KiB saved	✓
bzip2	1.7MiB (1699879)	81.06%	6.4845	173 ms	111	✓ 388KiB saved	✓
xz	1.7MiB (1727576)	82.38%	6.5902	469 ms	75	✓ 361KiB saved	✓
zstd	1.8MiB (1841432)	87.81%	7.0245	13 ms	10	✓ 250KiB saved	✓

File: G

Original Size: 2526752 bytes (2.5MiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	712KiB (728945)	28.85%	2.3079	97 ms	61	✓ 1.8MiB saved	✗
gzip	913KiB		2.9592	261 ms	18	✓	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
	(934659)	36.99%				1.6MiB saved	
bzip2	709KiB (725142)	28.70%	2.2959	199 ms	102	✓ 1.8MiB saved	✓
xz	734KiB (750608)	29.71%	2.3765	1003 ms	41	✓ 1.7MiB saved	✓
zstd	884KiB (905141)	35.82%	2.8658	27 ms	18	✓ 1.6MiB saved	✓

File: H

Original Size: 1022760 bytes (999KiB)

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
g07	437KiB (447485)	43.75%	3.5002	106 ms	99	✓ 562KiB saved	✗
gzip	440KiB (450169)	44.02%	3.5212	54 ms	11	✓ 560KiB saved	✓
bzip2	423KiB (432998)	42.34%	3.3869	71 ms	51	✓ 576KiB saved	✓
xz	359KiB (366916)	35.88%	2.8700	245 ms	24	✓ 641KiB	✓

Tool	Compressed Size	Ratio	Bits/ Symbol	Compress Time	Decompress Time	Space Saved	Lossless
						saved	
zstd	449KiB (458829)	44.86%	3.5889	25 ms	16	✓ 551KiB saved	✓

Performance Metrics

Compression Speed (Average)

Tool	Avg Compress Time	Ranking
zstd	19 ms	 #1
g07	76 ms	 #2
gzip	84 ms	 #3
bzip2	131 ms	#4
xz	432 ms	#5

Decompression Speed (Average)

Tool	Avg Decompress Time	Ranking
zstd	13 ms	 #1
gzip	14 ms	 #2
xz	36 ms	 #3
bzip2	77 ms	#4
g07	92 ms	#5

Rankings

Overall Compression Ratio Ranking

Rank	Tool	Average Ratio	Files Won	Total Compressed	Performance
 1	xz	54.16%	4	6.8MiB	★ ★ ★
 2	g07	54.80%	2	6.9MiB	★ ★
 3	bzip2	54.88%	2	6.9MiB	★
4	zstd	58.58%	0	7.4MiB	
5	gzip	59.47%	0	7.5MiB	

Files Won by Each Tool

- **g07**: 2 file(s)
- **gzip**: 0 file(s)
- **bzip2**: 2 file(s)
- **xz**: 4 file(s)
- **zstd**: 0 file(s)

Overall Statistics

Tool	Total Original	Total Compressed	Average Ratio	Bits/ Symbol	Files Tested
g07	13MiB	6.9MiB	54.80%	4.3836	8
gzip	13MiB	7.5MiB	59.47%	4.7579	8
bzip2	13MiB	6.9MiB	54.88%	4.3905	8

Tool	Total Original	Total Compressed	Average Ratio	Bits/ Symbol	Files Tested
xz	13MiB	6.8MiB	54.16%	4.3325	8
zstd	13MiB	7.4MiB	58.58%	4.6867	8

Report generated on Thu Mar 19 03:19:56 AM UTC 2026